DSA 210 Final Report
The Impact of the Academic Calendar and Physical Activity on Personal Biochemistry
Defne Akman | Student ID: 00032428
Spring 2025–2026

1. Motivation

I'm a student who checks her blood work whenever something feels off, I felt worse during the really stressful academic periods. But I couldn't tell if that was real. So I decided to actually look at the data and find out.

The point of this project is personal. I wanted to see if the stress from academics which are finals, failed courses, internships actually shows up in your blood data. And whether moving around more or less in a week actually makes a difference. The motivation was whether your own biochemistry is changing based on how busy and stressful you are.

Most stress researchs are done in groups, but when you track one person over time, you catch patterns. That is a point that group studies would completely miss. Your own data is way more detailed than in a regular study.

2. Data Sources

2.1 Blood Test Data from e-Nabız

I extracted my blood test results from e-Nabız (Turkey's national health record system) between September 2022 - July 2025. That's 12 separate test sessions across different hospitals. I didn't go on any regular schedule. I just got tested whenever I felt something was wrong, which means my data is probably biased toward sickness and stress.

For each test, I manually extracted the biomarkers into a spreadsheet.

Category	Parameters
Lipid panel	LDL, HDL, Total Cholesterol, Triglycerides
Glucose metabolism	Fasting Glucose, HbA1c, Fasting Insulin
Inflammation	CRP (C-reactive protein)
Thyroid	TSH, FT3, FT4, Anti-TPO
Iron & vitamins	Ferritin, Serum Iron, Iron Binding Capacity, Folate, Vitamin B12, Vitamin D, Zinc
Liver function	ALT, AST, ALP, GGT, Albumin
Kidney & metabolic	Creatinine, BUN, Uric Acid, Calcium, Phosphorus, Magnesium

Complete blood count	WBC, Hemoglobin, Hematocrit, RBC, MCV, MCH, MCHC, Platelets

2.2 Physical Activity Data from Apple Health

I exported my step count data from Apple Health app— that gave me about 78,000 records from March 2018 to April 2026. I took averages to daily totals, to 2,931 days of data. For each blood test, I calculated the average steps from the 14 days before and used that as a feature in the analysis.

2.3 Academic Calendar - Sabancı University

I used the official Sabancı academic calendar to label each blood test session by what period I was in:

- Label 0 Low/Holiday: before university or between semesters
- Label 1 Regular semester: normal lecture period, no extra stress
- Label 2 High stress: finals coming up or retaking courses
- Label 3 Extreme stress: failed courses + internships at the same time (Fall 2024-25 GPA 1.52; Summer 2024-25 with staj, MATH306 D, IE312 D; GPA 1.00)

3. Data Analysis

I approached this in four stages.

<u>Stage 1 - Data Collection & Preprocessing</u>

I went through the e-Nabız PDFs and manually pulled out the test values into a structured spreadsheet. Wrote code to parse the Apple Health XML. I had 12 blood test sessions total, but only 10 of them had enough biomarkers (at least 8 out of 12) to actually use in the machine learning part. For missing values, I filled them in with the median from that column.

<u>Stage 2 - Exploratory Data Analysis & Hypothesis Testing</u>

Three main hypotheses are tested:

H1: Blood biomarkers differ between normal periods and holidays

Mann-Whitney U test is used, because it's better for small sample sizes than a regular t-test.

Result: Nothing statistically significant (all p-values > 0.05), but there were some patterns. LDL and cholesterol were higher when I wasn't on holiday. HDL and ferritin seemed lower during the super stressful semesters.

H2: Step count correlates with blood markers

Pearson and Spearman correlations have been done between my 14-day average steps and the biomarkers.

Result: Fasting glucose had the strongest negative correlation ($r = -0.44$). TSH showed a negative correlation with 7-day step averages ($r = -0.49$). None of these reached the $p < 0.05$ threshold, but that's just because n=10 is small.

H3: I probably walk less during finals than during breaks
Mann-Whitney U is used comparing daily steps in finals windows vs. holidays (2022–2026).
Result: During finals I have average 4,527 steps/day, but 5,916 during holidays ($p = 0.078$). So we can say that stress really does make me move around less.

<u>Stage 3 — Full Biomarker Timeline Analysis</u>

I tracked all the biomarkers over time against what was happening academically. Some key things jumped out:

- CRP (inflammation marker): It was 1.7 mg/L before university. After that? It never went below 5 mg/L in any test (range: 5.8–13.87 mg/L). So basically, once university started, my inflammation just... stayed elevated. But its a biased information due to the time of the tests are only when ı feel something is off.

- TSH (thyroid): 5.05 mIU/L on January 2, 2024 which is totally out of range (should be 0.27–4.2). This was four days before Fall 2023-24 finals started. Weirdly, it was completely normal in my next test without any treatment.

- LDL and cholesterol: Peaked at 136 and 241 mg/dL in July 2025. That was the summer I had internship and got D's in MATH306 and IE312 (GPA 1.00 - literally my worst semester). Highest values the entire four years.

- Fasting insulin: This one was weird. My glucose was always normal (81.9–99.0 mg/dL), but my insulin was above the threshold in 3 out of 5 tests. HOMA-IR (a measure of insulin resistance) was over 2.5 in all 5 tests (should be below 2.5). So my pancreas was basically overcompensating normal glucose, but my body had to pump out more insulin to maintain it.

- Ferritin and folate: Both decreased progressively, especially during high-stress periods. Ferritin went from 41.77 ng/mL (December 2024) down to 15.0 ng/mL (July 2025). Folate decreased below reference in Spring 2025 (3.88) and Summer 2025 (4.12).

<u>Stage 4 — Machine Learning Methods</u>

I used 13 features in total: the 12 main biomarkers plus 14-day average step count. Everything was standardized (z-score) before running the models.

PCA (Dimensionality Reduction)
First main component with 33% of variance was by the cholesterol and inflammation markers. PC2 with 21% of variance was more about blood counts (hemoglobin, WBC, ferritin). The January 2024 test (TSH=5.05, CRP=12.4) showed up as this really isolated point, confirming it as a total biochemical outlier.

K-Means Clustering (k=3)
I tested different k values using silhouette score. k=3 gave the best score (0.140). Interesting part: without feeding the algorithm with any stress labels, it still put both my "Extreme stress" tests (December 2024 and July 2025) in the same cluster.

Binary Classification with LOO-CV
The tests was grouped into Low/Regular stress (labels 0–1, n=6) vs. High/Extreme stress (labels 2–3, n=4). Because n is so small, I used Leave-One-Out Cross-Validation instead of a normal train/test split. Tried Logistic Regression and Decision Trees. Accuracy was around 60% — basically the same as just guessing the majority class. That's expected when you've got 9 training samples and 13 features, so honestly, not surprising.

Random Forest Feature Importance
TSH (16%), CRP (13%), and average steps (11%) came out as the top three predictors. Ranked higher than the cholesterol markers. So for predicting stress, the thyroid-inflammation-activity pathway is more useful than just looking at lipids.

Time-Lag Spearman Correlation
Steps at 7, 14, 21, and 28 days before a test predicted biomarker values. Strongest finding was steps taken 7 days before vs. TSH came out to $r = -0.49$ ($p = 0.15$). Negative correlation means more active weeks means lower TSH. The effect size isn't small (it's medium-to-large), but it doesn't hit $p < 0.05$ just because of the small sample size.

4. Key Findings

<u>Finding 1 — I've been chronically inflamed since university started</u>
My CRP was 1.7 mg/L before I started at Sabancı. Since then? It's never dropped below 5 mg/L. That's a big jump. It stayed elevated across all 10 subsequent tests (5.8–13.87 mg/L). This suggests that university life itself just... keeps inflammation high, regardless of when I test. There's solid research showing that chronic stress activates pathways that increase inflammation (Cohen et al., 2012), so this makes sense mechanically.

<u>Finding 2 — My thyroid spiked right before finals</u>

TSH hit 5.05 mIU/L on January 2, 2024 (normal range is 0.27–4.2). It was the only out-of-range thyroid value in four years of testing. And it happened four days before Fall finals started. Next test? Completely normal, no treatment needed. So it was basically a stress response — there's literature on acute stress disrupting the hypothalamic-pituitary-thyroid axis (Tsigos & Chrousos, 2002), and this was a textbook example. The PCA analysis even confirmed this test as a biochemical outlier.

Finding 3 — Everything metabolically tanked in July 2025
July 2025 was when I had internship + the D's in MATH306 and IE312 (GPA 1.00). LDL increased up to 136 mg/dL (should be <100) and cholesterol to 241 mg/dL (should be <200). Both were the highest values in four years. Stress hormones like cortisol actually upregulate the enzyme that makes cholesterol, so it makes sense that my worst academic period matched my worst lipid profile (Rosmond, 2005).

Finding 4 — I have subclinical insulin resistance
Fasting glucose is always totally normal (81.9–99.0 mg/dL). But fasting insulin is above the reference range in 3 out of 5 tests. HOMA-IR was above 2.5 in all 5 tests which should be under 2.5. This pattern of normal glucose and high insulin means my pancreas has to work overtime just to keep blood sugar stable. That's a known precursor to type 2 diabetes and metabolic syndrome. Not great.

Finding 5 — Moving around more means better thyroid regulation
7-day step count negatively correlates with TSH ($r = -0.49$, $p = 0.15$). Translation: weeks when I walked more, my TSH was lower. This aligns with existing research showing that aerobic exercise suppresses TSH through dopamine pathways (Hackney, 2006). So the connection between activity and thyroid function seems real.

Finding 6 — Stress depletes micronutrients
Ferritin dropped from 41.77 ng/mL (December 2024) to 15.0 ng/mL (July 2025). Iron fell below reference range in that same July test. Folate went below reference in both Spring 2025 (3.88) and Summer 2025 (4.12) — my two highest-stress periods. Probably a combination of eating worse under stress and my body just demanding more from reserves.

5. Limitations and Future Work

Limitations
Small sample size (n=10 blood tests)
This is the biggest issue. At n=10, you need a correlation of $|r| > 0.63$ to hit $p < 0.05$. I'm finding effects around $r = 0.45–0.50$, which are real effect sizes but just can't reach significance with this few data points. Everything here is exploratory.

Selection bias in timing
I only got blood work when something felt wrong, not on a regular schedule. This systematically biases everything toward sickness and stress. I'm probably underrepresenting what my biochemistry looks like during calm periods.

Unmeasured confounders
Diet, sleep, menstrual cycle, season, whether I was actually sick — all of these mess with blood markers. None of them are controlled for. Some of these patterns might be driven by things other than academic stress.

Temporal leakage in step counts
The 14-day average I computed before each test partially overlaps with the stress period itself. So the correlation between steps and biomarkers might just be picking up the stress period, not predicting it.

Stress labeling is arbitrary
I split stress into four buckets based on grades and the calendar, but stress is actually continuous and subjective. The line between "high" and "extreme" is just... me deciding. A proper validated stress scale would be better.

Future Work
Regular blood tests on a fixed schedule
Next year, I'm going to get tested every 6 weeks no matter what, instead of only when I feel bad. That'll actually let me see the true patterns.

Get the sample size up to 30
I'd need about 29 tests to have proper statistical power. With regular testing, that's doable in two years.

Add sleep and HRV data
Apple Health has sleep duration and heart rate variability. Adding those would let me separate out whether low activity or bad sleep is the culprit.

Use an actual stress scale
Instead of guessing based on grades, I could fill out the Perceived Stress Scale (PSS-10) weekly. That's what researchers actually use.

Better statistical methods

With n this small, Bayesian methods probably make more sense than what I did. They let you build in prior knowledge about how biomarkers relate to stress.

Continuous glucose monitoring
Given the insulin resistance thing, a continuous glucose monitor would give way more information than a single fasting glucose number.

References

- Cohen, S., Janicki-Deverts, D., & Miller, G. E. (2012). Psychological stress and disease. JAMA, 298(14), 1685–1687.
- Hackney, A. C. (2006). Stress and the neuroendocrine system: the role of exercise as a stressor and modifier of stress. Expert Review of Endocrinology & Metabolism, 1(6), 783–792.
- Rosmond, R. (2005). Role of stress in the pathogenesis of the metabolic syndrome. Psychoneuroendocrinology, 30(1), 1–10.
- Tsigos, C., & Chrousos, G. P. (2002). Hypothalamic–pituitary–adrenal axis, neuroendocrine factors and stress. Journal of Psychosomatic Research, 53(4), 865–871.
- Arlot, S., & Celisse, A. (2010). A survey of cross-validation procedures for model selection. Statistics Surveys, 4, 40–79.
- Cohen, J. (1992). A power primer. Psychological Bulletin, 112(1), 155–159.

AI Usage Disclosure

This project utilizes AI assistance (Claude/Gemini) as permitted by the course guidelines. Below is the documentation of the AI interaction:

Prompt Used: "this is my project guideline. 'The Impact of the Academic Calendar and Physical Activity on Personal Biochemistry...' guide me through what you can do"

My contributions: The core and idea of this project is entirely mine. I designed the research question from personal curiosity about my own health patterns over four years of university. I collected all data myself — retrieving blood test records from e-Nabız across 12 sessions spanning 2022–2025, exporting my Apple Health data, and manually cross-referencing the Sabancı University academic calendar. I defined the stress labelling system (0–3) based on my own lived academic experience, including failed courses, repeated semesters, and internship overlap — context that no external tool could know. I verified every biological interpretation against my personal health history and made all decisions about which findings were meaningful.

AI assistance: Claude helped with the technical implementation: parsing the Apple Health XML, structuring the dataset, writing analysis and ML code, and formatting the report. All data values, stress labels, research framing, and interpretations are my own.